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SmishGuard: A Logistic Regression, Decision Tree, and SVM-Based Approach to Detecting Banking and Online Payment Smishing in Filipino/Taglish SMS Messages

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TRIOjan_Virus

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Chapter 1: Introduction

1.1 Project Context

SMS based phishing, known as smishing, includes fake delivery notifications, fake bank alerts, and prize winning messages with malicious links. It has become one of the most reported cybercrime types in the Philippines, especially through unregistered SIMs before the SIM Registration Act took effect. Most anti-spam filters are built for email and do not generalize well to the short, informal, code switched Tagalog English language used in local SMS scams.

1.2 Purpose and Description

The project aims to build a supervised classification model that labels an incoming SMS message as legitimate or smishing or scam based on text derived features such as the presence of shortened URLs, urgency keywords like "kailangan agad" or "claim now," sender ID patterns, presence of monetary or prize mentions, message length, and others. The model will be trained and evaluated using Logistic Regression, Decision Tree, and SVM, and compared on accuracy, precision, and recall. Recall matters more here because missing a real scam is worse than a false alarm.

1.3 Statement of the Problem

- How accurately can supervised ML models classify SMS messages as legitimate or smishing based on extracted text features?
- Which algorithm, Logistic Regression, Decision Tree, or SVM, performs best in terms of precision and recall for this task?
- Which features contribute most to correctly identifying a smishing message?

1.4 Hypothesis

The study is guided by the following hypotheses:

- Supervised machine learning models (Logistic Regression, Decision Tree, and SVM) can classify Filipino/Taglish SMS messages as legitimate or banking/online payment smishing with significantly higher accuracy than random guessing.
- There is a significant difference in performance among Logistic Regression, Decision Tree, and SVM in terms of precision and recall for detecting smishing messages.
- Text- and structure-based features (e.g., presence of shortened URLs, urgency keywords, monetary/prize mentions, sender ID patterns, message length) significantly contribute to the correct classification of smishing messages.
- Models optimized for higher recall will identify a greater proportion of actual smishing messages, at the cost of some increase in false positives, compared to models optimized primarily for accuracy.

These hypotheses will be tested through experimental evaluation using a labeled dataset of Filipino/Taglish SMS messages.

1.5 Research Objectives

1.5.1 General Objective

The general objective of this study is to develop and evaluate a supervised machine learning–based system that detects banking and online payment smishing in Filipino/Taglish SMS messages using Logistic Regression, Decision Tree, and Support Vector Machine (SVM) classifiers.

1.5.2 Specific Objectives

Specifically, the study aims to:

1. Collect and curate a labeled dataset of Filipino/Taglish SMS messages containing both legitimate and banking/online payment smishing instances.
2. Preprocess the SMS data using appropriate text cleaning, tokenization, and normalization techniques suitable for code-switched Filipino/English text.
3. Extract relevant text- and message-level features, including but not limited to:
    a. Presence of shortened or suspicious URLs
    b. Urgency-related keywords (e.g., “kailangan agad,” “claim now”)
    c. Monetary, prize, or account-related terms
    d. Sender ID patterns and phone number formats
    e. Message length and special character usage
4. Train three supervised classification models: Logistic Regression, Decision Tree, and SVM, using the extracted features.
5. Evaluate and compare the performance of the three models using metrics such as accuracy, precision, recall, and F1-score, with emphasis on recall due to the high cost of missing actual smishing messages.
6. Identify the most important features contributing to the correct classification of smishing messages using model-specific feature importance or coefficient analysis.
7. Provide recommendations on which algorithm and feature set are most suitable for deployment in a practical smishing detection system for the Philippine context.

1.6 Significance of the Study

This study addresses the growing threat of banking and online payment smishing in the Philippines by developing a machine learning–based detection approach tailored to Filipino and Taglish SMS messages. As mobile banking and digital wallet usage expand, scammers increasingly exploit informal, code-switched language patterns that many existing English-centric filters fail to capture. By focusing on locally relevant text and message-level features, this research aims to improve the accuracy and practicality of smishing detection in the Philippine context. The results of this study are expected to benefit the following stakeholders:

1.6.1 Filipino Mobile Phone Users

The main beneficiaries are mobile phone users in the Philippines who may be targeted by banking and online payment scams through SMS. A reliable detection model can help protect users from financial loss and identity theft by identifying suspicious messages before they click harmful links or share personal information.

1.6.2 Banks and Online Payment Service Providers

Banks and e-wallet companies in the Philippines can use the study's findings to improve security warnings and develop smishing filters that recognize local language patterns.

1.6.3 Telecommunications Companies

Local telcos can use the study's findings to improve SMS spam filters, especially for Taglish messages that may be harder for systems designed mainly for English to detect.

1.6.4 Cybersecurity Researchers and the Academic Community

This study adds to research on SMS phishing detection by focusing on Filipino and Taglish messages. Its findings can also help future studies on smishing detection for other Southeast Asian languages.

1.6.5 Regulatory and Law Enforcement Agencies.

The study can support the goals of the SIM Registration Act and other cybersecurity policies by providing information about common smishing messages in the Philippines. This can help improve enforcement and public awareness campaigns.

1.6.6 Future Researchers

This study can serve as a starting point for future research using larger datasets, more Philippine languages, or advanced deep learning methods.

1.7 Scope and Limitations

This study builds and tests machine learning models that detect smishing about online payments and banking in Filipino, English, or Taglish messages. The system can tell if a message is smishing or legitimate by looking at the signs in the message. For example, it looks at shortened links, words that create a sense of urgency, mentions of money or prizes, sender ID patterns, and the length of the message. It uses three methods: Logistic Regression, Decision Tree, and Support Vector Machine. It compares how well each one works in terms of their accuracy, precision, and recall scores. However, the system only reads the message itself. It can’t check if a link is actually harmful, confirm who the sender is with the phone companies, or check pictures, attachments, and scam phone calls. It also can’t detect other kinds of scams or general spam, and it doesn’t work for other Philippine languages or dialects. The results will depend on

the number and quality of the dataset. Real text messages are hard to get because of privacy, so the dataset might not completely match the messages people actually receive. The study doesn't use deep learning or transformer models. It's also only a proof of concept, so it is not used as a live filter and it is not connected to any phone company or bank system. The models only know the patterns in the dataset and might need to be trained again to catch new scam wording, since scammers are always changing how they trick and manipulate people.

Chapter 2: Review of Related Literature and Studies

2.1 SMS Smishing and Financial Scams

Smishing, or SMS phishing, is a form of social engineering in which attackers use text messages to deceive recipients into clicking malicious links, providing personal information, or performing unauthorized actions. Banking and online payment services are common targets because scammers may use fake account alerts, transaction notices, payment requests, and reward claims to create a sense of urgency. Mishra and Soni (2020) emphasized the importance of analyzing SMS content and URL behavior in detecting smishing, while Mir et al. (2026) identified suspicious links, urgent language, account-related warnings, and financial requests as common characteristics of SMS phishing. Samad et al. (2023) also demonstrated the use of natural language processing and machine learning for detecting smishing messages.

2.2 Banking and Online Payment Smishing Indicators

Several characteristics can serve as indicators for identifying banking and online payment-related smishing messages. These include suspicious or shortened URLs, urgent words or phrases, requests to verify or update an account, financial references, reward or prize claims, phone numbers, unusual sender information, and special characters. Mishra and Soni (2022) identified features such as URL presence, phone numbers, smishing keywords, symbols, and special characters as useful indicators for classification. Mishra and Soni (2020) also emphasized the importance of analyzing message content and URLs. Kohilan et al. (2023) further explored machine learning-based detection of smishing at the end-user level. These findings provide a basis for selecting relevant features for detecting banking and online payment smishing messages.

2.3 Philippine Context of Smishing

Smishing is a relevant cybersecurity concern in the Philippines, particularly in relation to mobile communication and digital transactions. Diaz and Balbacal (2024) examined Filipino attitudes and perceptions toward the SIM Registration Act, providing context on regulatory efforts related to mobile communication and security. Blancaflor et al. (2024) examined smishing as a social engineering technique used to gain access to phone contacts. Local studies have also explored machine learning-based SMS scam detection. Corpuz et al. (2026) developed a machine learning approach for SMS phishing in the Philippine context, while Parker et al. (2025) developed BantayText for detecting scam SMS messages in the Philippines. These studies demonstrate the importance of developing detection approaches suited to the local SMS environment.

2.4 Filipino and Taglish SMS Messages

Filipino SMS messages may contain informal expressions, abbreviations, and combinations of Filipino and English, which can create challenges for text classification. Montalan et al. (2025) introduced Batayan, a Filipino NLP benchmark that highlights the need for resources designed for Filipino language processing. De Leon et al. (2025) explored a Philippine-based anti-scam SMS relay system using few-shot LaBSE, demonstrating the application of language models to local scam detection. Corpuz et al. (2026) specifically considered Filipino and code-switched SMS messages. Similarly, Ghourabi et al. (2020) demonstrated the application of deep learning to SMS spam detection involving Arabic and English messages. These studies support the importance of considering multilingual and code-switched characteristics when developing SMS classification systems.

2.5 Machine Learning for SMS and Smishing Detection

Machine learning has been widely applied to SMS classification using textual and structural features. De Luna et al. (2023) demonstrated the use of machine learning for SMS spam detection, while Palad et al. (2020) evaluated decision tree classification algorithms using fraud-related datasets. Soysaldı Şahin et al. (2026) further showed that feature representation can influence the performance of classical machine learning models. Corpuz et al. (2026) used TF-IDF and sender information and compared several algorithms, including Support Vector Machine, Logistic Regression, Random Forest, K-Nearest Neighbors, and Multinomial Naive Bayes. These studies support the use of text-based and message-level features as inputs for supervised SMS classification.

2.6 Deep Learning and Advanced Detection Approaches

Recent studies have also examined deep learning and transformer-based methods for detecting malicious or phishing-related messages. Munoz and Islam (2025) investigated deep learning approaches for multi-class phishing text classification. Uddin et al. (2025) developed an explainable transformer-based approach for SMS spam detection, demonstrating the potential of advanced language models while emphasizing interpretability. Ghourabi et al. (2020) applied a hybrid CNN-LSTM model to multilingual SMS spam detection, while Samad et al. (2023) combined natural language processing and machine learning for smishing detection. These studies demonstrate that both traditional machine learning and advanced deep learning approaches can be applied to SMS security.

2.7 Classification Algorithms

2.7.1 Logistic Regression

Logistic Regression is a supervised classification algorithm commonly used for binary classification. In the proposed study, it can be used to classify SMS messages as either legitimate or smishing based on extracted features. Its relative simplicity and interpretability allow the contribution of individual features to be examined through the model's learned coefficients.

2.7.2 Decision Tree

Decision Tree classification uses a sequence of decision rules based on input features. For example, a model may determine whether combinations of URLs, urgency terms, and financial keywords increase the likelihood that a message is smishing. Palad et al. (2020) demonstrated the use of decision tree classification algorithms in fraud-related datasets, supporting their applicability to classification problems involving potentially fraudulent activities.

2.7.3 Support Vector Machine

Support Vector Machine (SVM) is a supervised learning algorithm that identifies a decision boundary for separating different classes. It is commonly applied to text classification because it can perform effectively with high-dimensional feature representations such as TF-IDF. Corpuz et al. (2026) reported strong performance from SVM in Philippine SMS phishing detection. Therefore, SVM provides an important model for comparison with Logistic Regression and Decision Tree in the proposed study.

2.8 Synthesis and Research Gap

The reviewed literature and studies show that smishing messages can be identified through indicators such as suspicious URLs, urgent language, financial references, account-related requests, sender information, and other message characteristics. Previous research has also demonstrated that machine learning, natural language processing, and deep learning can be applied to SMS spam and smishing detection.

However, many existing studies focus on general SMS spam, phishing, or datasets from other languages and contexts. There is still a need for approaches that specifically address Filipino/Taglish banking and online payment-related SMS messages, where local language patterns and context may affect classification. Although Corpuz et al. (2026) addressed SMS phishing in the Philippine context, the proposed study further focuses specifically on banking and online payment-related messages and examines the contribution of relevant smishing indicators.

Therefore, this study proposes a supervised machine learning approach for detecting banking and online payment smishing in the Philippine context. It will examine relevant message indicators and compare Logistic Regression, Decision Tree, and Support Vector Machine based on their classification performance. The study also aims to identify which features contribute most to the accurate classification of legitimate and smishing SMS messages.

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